STA2002 Probability and Statistics II

Mid-term Examination

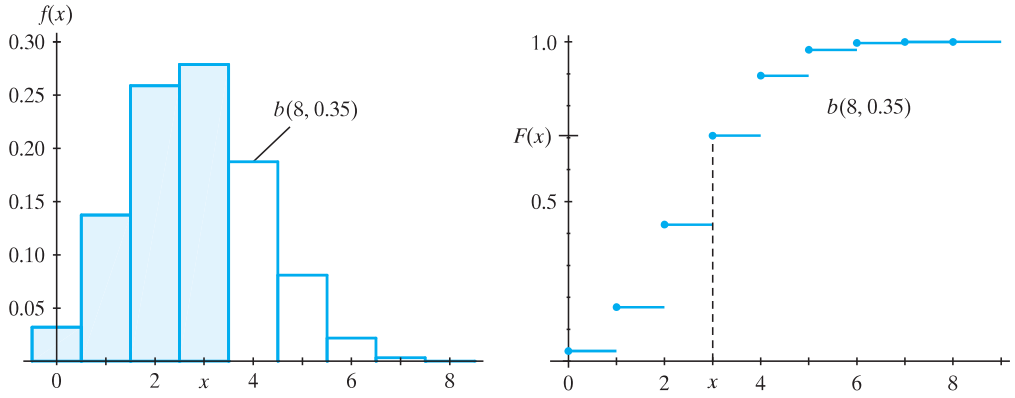
October 27, 2019: 9:00am - 11:30am

Name: _____ Student ID: _____

Answer ALL questions in the Answer Book.

Question	Points	Score
1	4	
2	4	
3	4	
4	4	
5	4	
6	4	
7	4	
8	12	
9	12	
10	12	
11	20	
12	16	
Total:	100	

Table II The Binomial Distribution



$$F(x) = P(X \le x) = \sum_{k=0}^{x} \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
2	0	0.9025	0.8100	0.7225	0.6400	0.5625	0.4900	0.4225	0.3600	0.3025	0.2500
	1	0.9975	0.9900	0.9775	0.9600	0.9375	0.9100	0.8775	0.8400	0.7975	0.7500
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.8574	0.7290	0.6141	0.5120	0.4219	0.3430	0.2746	0.2160	0.1664	0.1250
	1	0.9928	0.9720	0.9392	0.8960	0.8438	0.7840	0.7182	0.6480	0.5748	0.5000
	2	0.9999	0.9990	0.9966	0.9920	0.9844	0.9730	0.9571	0.9360	0.9089	0.8750
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.8145	0.6561	0.5220	0.4096	0.3164	0.2401	0.1785	0.1296	0.0915	0.0625
	1	0.9860	0.9477	0.8905	0.8192	0.7383	0.6517	0.5630	0.4752	0.3910	0.3125
	2	0.9995	0.9963	0.9880	0.9728	0.9492	0.9163	0.8735	0.8208	0.7585	0.6875
	3	1.0000	0.9999	0.9995	0.9984	0.9961	0.9919	0.9850	0.9744	0.9590	0.9375
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0	0.7738	0.5905	0.4437	0.3277	0.2373	0.1681	0.1160	0.0778	0.0503	0.0312
	1	0.9774	0.9185	0.8352	0.7373	0.6328	0.5282	0.4284	0.3370	0.2562	0.1875
	2	0.9988	0.9914	0.9734	0.9421	0.8965	0.8369	0.7648	0.6826	0.5931	0.5000
	3	1.0000	0.9995	0.9978	0.9933	0.9844	0.9692	0.9460	0.9130	0.8688	0.8125
	4	1.0000	1.0000	0.9999	0.9997	0.9990	0.9976	0.9947	0.9898	0.9815	0.9688
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	0	0.7351	0.5314	0.3771	0.2621	0.1780	0.1176	0.0754	0.0467	0.0277	0.0156
	1	0.9672	0.8857	0.7765	0.6553	0.5339	0.4202	0.3191	0.2333	0.1636	0.1094
	2	0.9978	0.9842	0.9527	0.9011	0.8306	0.7443	0.6471	0.5443	0.4415	0.3438
	3	0.9999	0.9987	0.9941	0.9830	0.9624	0.9295	0.8826	0.8208	0.7447	0.6562
	4	1.0000	0.9999	0.9996	0.9984	0.9954	0.9891	0.9777	0.9590	0.9308	0.8906
	5	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9982	0.9959	0.9917	0.9844
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	0	0.6983	0.4783	0.3206	0.2097	0.1335	0.0824	0.0490	0.0280	0.0152	0.0078
	1	0.9556	0.8503	0.7166	0.5767	0.4449	0.3294	0.2338	0.1586	0.1024	0.0625

Table II *continued*

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
	2	0.9962	0.9743	0.9262	0.8520	0.7564	0.6471	0.5323	0.4199	0.3164	0.2266
	3	0.9998	0.9973	0.9879	0.9667	0.9294	0.8740	0.8002	0.7102	0.6083	0.5000
	4	1.0000	0.9998	0.9988	0.9953	0.9871	0.9712	0.9444	0.9037	0.8471	0.7734
	5	1.0000	1.0000	0.9999	0.9996	0.9987	0.9962	0.9910	0.9812	0.9643	0.9375
	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9984	0.9963	0.9922
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	0	0.6634	0.4305	0.2725	0.1678	0.1001	0.0576	0.0319	0.0168	0.0084	0.0039
	1	0.9428	0.8131	0.6572	0.5033	0.3671	0.2553	0.1691	0.1064	0.0632	0.0352
	2	0.9942	0.9619	0.8948	0.7969	0.6785	0.5518	0.4278	0.3154	0.2201	0.1445
	3	0.9996	0.9950	0.9786	0.9437	0.8862	0.8059	0.7064	0.5941	0.4770	0.3633
	4	1.0000	0.9996	0.9971	0.9896	0.9727	0.9420	0.8939	0.8263	0.7396	0.6367
	5	1.0000	1.0000	0.9998	0.9988	0.9958	0.9887	0.9747	0.9502	0.9115	0.8555
	6	1.0000	1.0000	1.0000	0.9999	0.9996	0.9987	0.9964	0.9915	0.9819	0.9648
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9983	0.9961
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0	0.6302	0.3874	0.2316	0.1342	0.0751	0.0404	0.0207	0.0101	0.0046	0.0020
	1	0.9288	0.7748	0.5995	0.4362	0.3003	0.1960	0.1211	0.0705	0.0385	0.0195
	2	0.9916	0.9470	0.8591	0.7382	0.6007	0.4628	0.3373	0.2318	0.1495	0.0898
	3	0.9994	0.9917	0.9661	0.9144	0.8343	0.7297	0.6089	0.4826	0.3614	0.2539
	4	1.0000	0.9991	0.9944	0.9804	0.9511	0.9012	0.8283	0.7334	0.6214	0.5000
	5	1.0000	0.9999	0.9994	0.9969	0.9900	0.9747	0.9464	0.9006	0.8342	0.7461
	6	1.0000	1.0000	1.0000	0.9997	0.9987	0.9957	0.9888	0.9750	0.9502	0.9102
	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9986	0.9962	0.9909	0.9805
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992	0.9980
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
10	0	0.5987	0.3487	0.1969	0.1074	0.0563	0.0282	0.0135	0.0060	0.0025	0.0010
	1	0.9139	0.7361	0.5443	0.3758	0.2440	0.1493	0.0860	0.0464	0.0233	0.0107
	2	0.9885	0.9298	0.8202	0.6778	0.5256	0.3828	0.2616	0.1673	0.0996	0.0547
	3	0.9990	0.9872	0.9500	0.8791	0.7759	0.6496	0.5138	0.3823	0.2660	0.1719
	4	0.9999	0.9984	0.9901	0.9672	0.9219	0.8497	0.7515	0.6331	0.5044	0.3770
	5	1.0000	0.9999	0.9986	0.9936	0.9803	0.9527	0.9051	0.8338	0.7384	0.6230
	6	1.0000	1.0000	0.9999	0.9991	0.9965	0.9894	0.9740	0.9452	0.8980	0.8281
	7	1.0000	1.0000	1.0000	0.9999	0.9996	0.9984	0.9952	0.9877	0.9726	0.9453
	8	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995	0.9983	0.9955	0.9893
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9990
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
11	0	0.5688	0.3138	0.1673	0.0859	0.0422	0.0198	0.0088	0.0036	0.0014	0.0005
	1	0.8981	0.6974	0.4922	0.3221	0.1971	0.1130	0.0606	0.0302	0.0139	0.0059
	2	0.9848	0.9104	0.7788	0.6174	0.4552	0.3127	0.2001	0.1189	0.0652	0.0327
	3	0.9984	0.9815	0.9306	0.8389	0.7133	0.5696	0.4256	0.2963	0.1911	0.1133
	4	0.9999	0.9972	0.9841	0.9496	0.8854	0.7897	0.6683	0.5328	0.3971	0.2744
	5	1.0000	0.9997	0.9973	0.9883	0.9657	0.9218	0.8513	0.7535	0.6331	0.5000
	6	1.0000	1.0000	0.9997	0.9980	0.9924	0.9784	0.9499	0.9006	0.8262	0.7256

Table II *continued*

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
	7	1.0000	1.0000	1.0000	0.9998	0.9988	0.9957	0.9878	0.9707	0.9390	0.8867
	8	1.0000	1.0000	1.0000	1.0000	0.9999	0.9994	0.9980	0.9941	0.9852	0.9673
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9993	0.9978	0.9941
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9995
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
12	0	0.5404	0.2824	0.1422	0.0687	0.0317	0.0138	0.0057	0.0022	0.0008	0.0002
	1	0.8816	0.6590	0.4435	0.2749	0.1584	0.0850	0.0424	0.0196	0.0083	0.0032
	2	0.9804	0.8891	0.7358	0.5583	0.3907	0.2528	0.1513	0.0834	0.0421	0.0193
	3	0.9978	0.9744	0.9078	0.7946	0.6488	0.4925	0.3467	0.2253	0.1345	0.0730
	4	0.9998	0.9957	0.9761	0.9274	0.8424	0.7237	0.5833	0.4382	0.3044	0.1938
	5	1.0000	0.9995	0.9954	0.9806	0.9456	0.8822	0.7873	0.6652	0.5269	0.3872
	6	1.0000	0.9999	0.9993	0.9961	0.9857	0.9614	0.9154	0.8418	0.7393	0.6128
	7	1.0000	1.0000	0.9999	0.9994	0.9972	0.9905	0.9745	0.9427	0.8883	0.8062
	8	1.0000	1.0000	1.0000	0.9999	0.9996	0.9983	0.9944	0.9847	0.9644	0.9270
	9	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9992	0.9972	0.9921	0.9807
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9989	0.9968
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13	0	0.5133	0.2542	0.1209	0.0550	0.0238	0.0097	0.0037	0.0013	0.0004	0.0001
	1	0.8646	0.6213	0.3983	0.2336	0.1267	0.0637	0.0296	0.0126	0.0049	0.0017
	2	0.9755	0.8661	0.6920	0.5017	0.3326	0.2025	0.1132	0.0579	0.0269	0.0112
	3	0.9969	0.9658	0.8820	0.7473	0.5843	0.4206	0.2783	0.1686	0.0929	0.0461
	4	0.9997	0.9935	0.9658	0.9009	0.7940	0.6543	0.5005	0.3530	0.2279	0.1334
	5	1.0000	0.9991	0.9924	0.9700	0.9198	0.8346	0.7159	0.5744	0.4268	0.2905
	6	1.0000	0.9999	0.9987	0.9930	0.9757	0.9376	0.8705	0.7712	0.6437	0.5000
	7	1.0000	1.0000	0.9998	0.9988	0.9944	0.9818	0,9538	0.9023	0.8212	0.7095
	8	1.0000	1.0000	1.0000	0.9998	0.9990	0.9960	0.9874	0.9679	0.9302	0.8666
	9	1.0000	1.0000	1.0000	1.0000	0.9999	0.9993	0.9975	0.9922	0.9797	0.9539
	10	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9987	0.9959	0.9888
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995	0.9983
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
14	0	0.4877	0.2288	0.1028	0.0440	0.0178	0.0068	0.0024	0.0008	0.0002	0.0001
	1	0.8470	0.5846	0.3567	0.1979	0.1010	0.0475	0.0205	0.0081	0.0029	0.0009
	2	0.9699	0.8416	0.6479	0.4481	0.2811	0.1608	0.0839	0.0398	0.0170	0.0065
	3	0.9958	0.9559	0.8535	0.6982	0.5213	0.3552	0.2205	0.1243	0.0632	0.0287
	4	0.9996	0.9908	0.9533	0.8702	0.7415	0.5842	0.4227	0.2793	0.1672	0.0898
	5	1.0000	0.9985	0.9885	0.9561	0.8883	0.7805	0.6405	0.4859	0.3373	0.2120
	6	1.0000	0.9998	0.9978	0.9884	0.9617	0.9067	0.8164	0.6925	0.5461	0.3953
	7	1.0000	1.0000	0.9997	0.9976	0.9897	0.9685	0.9247	0.8499	0.7414	0.6047
	8	1.0000	1.0000	1.0000	0.9996	0.9978	0.9917	0.9757	0.9417	0.8811	0.7880
	9	1.0000	1.0000	1.0000	1.0000	0.9997	0.9983	0.9940	0.9825	0.9574	0.9102
	10	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9989	0.9961	0.9886	0.9713

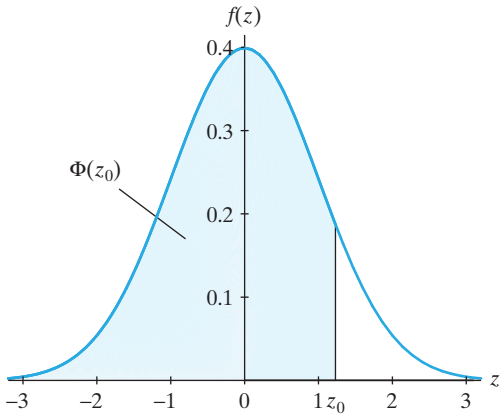
Table II *continued*

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9994	0.9978	0.9935
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9991
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
15	0	0.4633	0.2059	0.0874	0.0352	0.0134	0.0047	0.0016	0.0005	0.0001	0.0000
	1	0.8290	0.5490	0.3186	0.1671	0.0802	0.0353	0.0142	0.0052	0.0017	0.0005
	2	0.9638	0.8159	0.6042	0.3980	0.2361	0.1268	0.0617	0.0271	0.0107	0.0037
	3	0.9945	0.9444	0.8227	0.6482	0.4613	0.2969	0.1727	0.0905	0.0424	0.0176
	4	0.9994	0.9873	0.9383	0.8358	0.6865	0.5155	0.3519	0.2173	0.1204	0.0592
	5	0.9999	0.9978	0.9832	0.9389	0.8516	0.7216	0.5643	0.4032	0.2608	0.1509
	6	1.0000	0.9997	0.9964	0.9819	0.9434	0.8689	0.7548	0.6098	0.4522	0.3036
	7	1.0000	1.0000	0.9994	0.9958	0.9827	0.9500	0.8868	0.7869	0.6535	0.5000
	8	1.0000	1.0000	0.9999	0.9992	0.9958	0.9848	0.9578	0.9050	0.8182	0.6964
	9	1.0000	1.0000	1.0000	0.9999	0.9992	0.9963	0.9876	0.9662	0.9231	0.8491
	10	1.0000	1.0000	1.0000	1.0000	0.9999	0.9993	0.9972	0.9907	0.9745	0.9408
	11	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995	0.9981	0.9937	0.9824
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9987	0.9989	0.9963
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
16	0	0.4401	0.1853	0.0743	0.0281	0.0100	0.0033	0.0010	0.0003	0.0001	0.0000
	1	0.8108	0.5147	0.2839	0.1407	0.0635	0.0261	0.0098	0.0033	0.0010	0.0003
	2	0.9571	0.7892	0.5614	0.3518	0.1971	0.0994	0.0451	0.0183	0.0066	0.0021
	3	0.9930	0.9316	0.7899	0.5981	0.4050	0.2459	0.1339	0,0651	0.0281	0.0106
	4	0.9991	0.9830	0.9209	0.7982	0.6302	0.4499	0.2892	0.1666	0.0853	0.0384
	5	0.9999	0.9967	0.9765	0.9183	0.8103	0.6598	0.4900	0.3288	0.1976	0.1051
	6	1.0000	0.9995	0.9944	0.9733	0.9204	0.8247	0.6881	0.5272	0.3660	0.2272
	7	1.0000	0.9999	0.9989	0.9930	0.9729	0.9256	0.8406	0.7161	0.5629	0.4018
	8	1.0000	1.0000	0.9998	0.9985	0.9925	0.9743	0.9329	0.8577	0.7441	0.5982
	9	1.0000	1.0000	1.0000	0.9998	0.9984	0.9929	0.9771	0.9417	0.8759	0.7728
	10	1.0000	1.0000	1.0000	1.0000	0.9997	0.9984	0.9938	0.9809	0.9514	0.8949
	11	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9987	0.9951	0.9851	0.9616
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9991	0.9965	0.9894
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9994	0.9979
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
20	0	0.3585	0.1216	0.0388	0.0115	0.0032	0.0008	0.0002	0.0000	0.0000	0.0000
	1	0.7358	0.3917	0.1756	0.0692	0.0243	0.0076	0.0021	0.0005	0.0001	0.0000
	2	0.9245	0.6769	0.4049	0.2061	0.0913	0.0355	0.0121	0.0036	0.0009	0.0002
	3	0.9841	0.8670	0.6477	0.4114	0.2252	0.1071	0.0444	0.0160	0.0049	0.0013
	4	0.9974	0.9568	0.8298	0.6296	0.4148	0.2375	0.1182	0.0510	0.0189	0.0059

Table II *continued*

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
	5	0.9997	0.9887	0.9327	0.8042	0.6172	0.4164	0.2454	0.1256	0.0553	0.0207
	6	1.0000	0.9976	0.9781	0.9133	0.7858	0.6080	0.4166	0.2500	0.1299	0.0577
	7	1.0000	0.9996	0.9941	0.9679	0.8982	0.7723	0.6010	0.4159	0.2520	0.1316
	8	1.0000	0.9999	0.9987	0.9900	0.9591	0.8867	0.7624	0.5956	0.4143	0.2517
	9	1.0000	1.0000	0.9998	0.9974	0.9861	0.9520	0.8782	0.7553	0.5914	0.4119
	10	1.0000	1.0000	1.0000	0.9994	0.9961	0.9829	0.9468	0.8725	0.7507	0.5881
	11	1.0000	1.0000	1.0000	0.9999	0.9991	0.9949	0.9804	0.9435	0.8692	0.7483
	12	1.0000	1.0000	1.0000	1.0000	0.9998	0.9987	0.9940	0.9790	0.9420	0.8684
	13	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9985	0.9935	0.9786	0.9423
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9984	0.9936	0.9793
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9985	0.9941
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9987
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
25	0	0.2774	0.0718	0.0172	0.0038	0.0008	0.0001	0.0000	0.0000	0.0000	0.0000
	1	0.6424	0.2712	0.0931	0.0274	0.0070	0.0016	0.0003	0.0001	0.0000	0.0000
	2	0.8729	0.5371	0.2537	0.0982	0.0321	0.0090	0.0021	0.0004	0.0001	0.0000
	3	0.9659	0.7636	0.4711	0.2340	0.0962	0.0332	0.0097	0.0024	0.0005	0.0001
	4	0.9928	0.9020	0.6821	0.4207	0.2137	0.0905	0.0320	0.0095	0.0023	0.0005
	5	0.9988	0.9666	0.8385	0.6167	0.3783	0.1935	0.0826	0.0294	0.0086	0.0020
	6	0.9998	0.9905	0.9305	0.7800	0.5611	0.3407	0.1734	0.0736	0.0258	0.0073
	7	1.0000	0.9977	0.9745	0.8909	0.7265	0.5118	0.3061	0.1536	0.0639	0.0216
	8	1.0000	0.9995	0.9920	0.9532	0.8506	0.6769	0.4668	0.2735	0.1340	0.0539
	9	1.0000	0.9999	0.9979	0.9827	0.9287	0.8106	0.6303	0.4246	0.2424	0.1148
	10	1.0000	1.0000	0.9995	0.9944	0.9703	0.9022	0.7712	0.5858	0.3843	0.2122
	11	1.0000	1.0000	0.9999	0.9985	0.9893	0.9558	0.8746	0.7323	0.5426	0.3450
	12	1.0000	1.0000	1.0000	0.9996	0.9966	0.9825	0.9396	0.8462	0.6937	0.5000
	13	1.0000	1.0000	1.0000	0.9999	0.9991	0.9940	0.9745	0.9222	0.8173	0.6550
	14	1.0000	1.0000	1,0000	1.0000	0.9998	0.9982	0.9907	0.9656	0.9040	0.7878
	15	1.0000	1.0000	1.0000	1.0000	1.0000	0.9995	0.9971	0.9868	0.9560	0.8852
	16	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9992	0.9957	0.9826	0.9461
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9988	0.9942	0.9784
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9984	0.9927
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9980
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table Va The Standard Normal Distribution Function



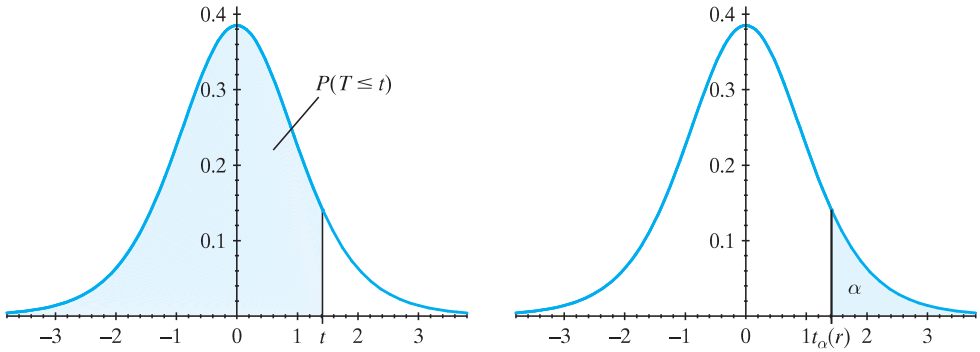
$$P(Z \le z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291

Table VI The t Distribution



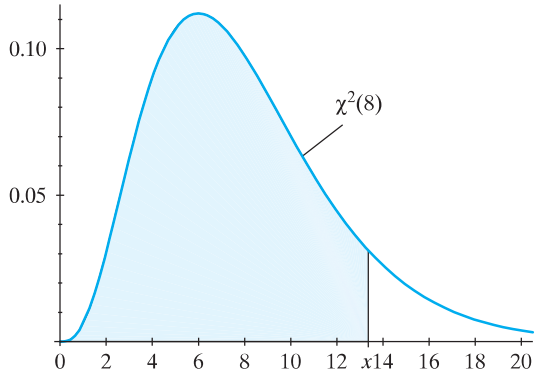
$$P(T \le t) = \int_{-\infty}^{t} \frac{\Gamma[(r+1)/2]}{\sqrt{\pi r}\,\Gamma(r/2)(1+w^2/r)^{(r+1)/2}}\,dw$$

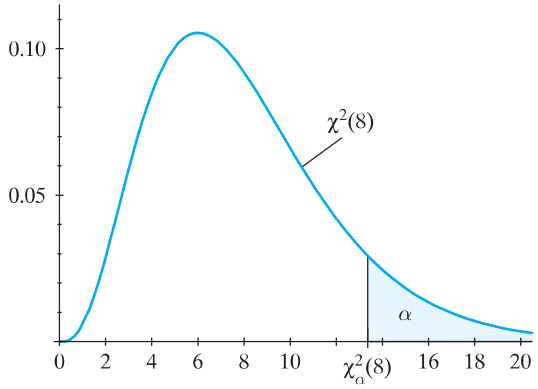
$$P(T \le -t) = 1 - P(T \le t)$$

	$P(T \le t)$						
	0.60	0.75	0.90	0.95	0.975	0.99	0.995
r	$t_{0.40}(r)$	$t_{0.25}(r)$	$t_{0.10}(r)$	$t_{0.05}(r)$	$t_{0.025}(r)$	$t_{0.01}(r)$	$t_{0.005}(r)$
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012
14	0.258	0.692	1.345	1.761	2.145	2.624	2.997
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576

This table is taken from Table III of Fisher and Yates: *Statistical Tables for Biological, Agricultrual, and Medical Research*, published by Longman Group Ltd., London (previously published by Oliver and Boyd, Edinburgh).

Table IV The Chi-Square Distribution





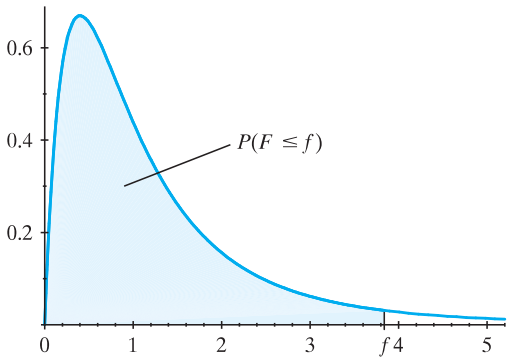
$$P(X \le x) = \int_0^x \frac{1}{\Gamma(r/2)2^{r/2}} w^{r/2-1} e^{-w/2} dw$$

	$P(X \le x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09
6	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81
7	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48
8	1.646	2.180	2.733	3.490	13.36	15.51	17.54	20.09
9	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67
10	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21
11	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72
12	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22
13	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69
14	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14
15	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58
16	5.812	6.908	7.962	9.312	23.54	26.30	28.84	32.00
17	6.408	7.564	8.672	10.08	24.77	27.59	30.19	33.41
18	7.015	8.231	9.390	10.86	25.99	28.87	31.53	34.80
19	7.633	8.907	10.12	11.65	27.20	30.14	32.85	36.19
20	8.260	9.591	10.85	12.44	28.41	31.41	34.17	37.57
21	8.897	10.28	11.59	13.24	29.62	32.67	35.48	38.93
22	9.542	10.98	12.34	14.04	30.81	33.92	36.78	40.29
23	10.20	11.69	13.09	14.85	32.01	35.17	38.08	41.64
24	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
25	11.52	13.12	14.61	16.47	34.38	37.65	40.65	44.31
26	12.20	13.84	15.38	17.29	35.56	38.88	41.92	45.64
27	12.88	14.57	16.15	18.11	36.74	40.11	43.19	46.96
28	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
29	14.26	16.05	17.71	19.77	39.09	42.56	45.72	49.59
30	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
40	22.16	24.43	26.51	29.05	51.80	55.76	59.34	63.69
50	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
60	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
70	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.4
80	53.34	57.15	60.39	64.28	96.58	101.9	106.6	112.3

This table is abridged and adapted from Table III in *Biometrika Tables for Statisticians*, edited by E.S.Pearson and H.O.Hartley.

Table VII The F Distribution

$$P(F \le f) = \int_0^f \frac{\Gamma[(r_1 + r_2)/2](r_1/r_2)^{r_1/2} w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)(1 + r_1 w/r_2)^{(r_1+r_2)/2}}\, dw$$



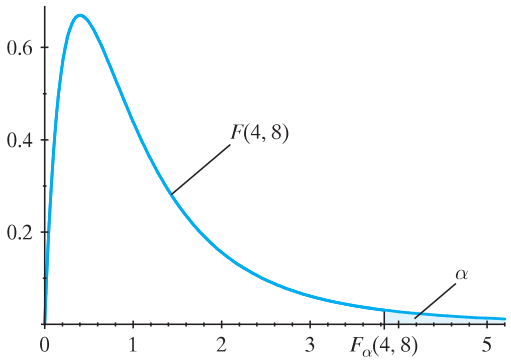


Table VII *continued*

$$P(F \le f) = \int_0^f \frac{\Gamma[(r_1 + r_2)/2](r_1/r_2)^{r_1/2} w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)(1 + r_1 w/r_2)^{(r_1+r_2)/2}}\, dw$$

α	$P(F \le f)$	Den. d.f. r_2	Numerator Degrees of Freedom, r_1									
			1	2	3	4	5	6	7	8	9	10
0.05	0.95	1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9
0.025	0.975		647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28	968.63
0.01	0.99		4052	4999.5	5403	5625	5764	5859	5928	5981	6022	6056
0.05	0.95	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
0.025	0.975		38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40
0.01	0.99		98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40
0.05	0.95	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
0.025	0.975		17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42
0.01	0.99		34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23
0.05	0.95	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
0.025	0.975		12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84
0.01	0.99		21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55
0.05	0.95	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
0.025	0.975		10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62
0.01	0.99		16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05
0.05	0.95	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
0.025	0.975		8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46
0.01	0.99		13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87
0.05	0.95	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
0.025	0.975		8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76
0.01	0.99		12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
0.05	0.95	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
0.025	0.975		7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
0.01	0.99		11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
0.05	0.95	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
0.025	0.975		7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
0.01	0.99		10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
0.05	0.95	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
0.025	0.975		6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72
0.01	0.99		10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85

Table VII *continued*

$$P(F \le f) = \int_0^f \frac{\Gamma[(r_1 + r_2)/2](r_1/r_2)^{r_1/2} w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)(1 + r_1 w/r_2)^{(r_1+r_2)/2}}\, dw$$

α	$P(F \le f)$	Den. d.f. r_2	Numerator Degrees of Freedom, r_1									
			1	2	3	4	5	6	7	8	9	10
0.05	0.95	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
0.025	0.975		6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37
0.01	0.99		9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30
0.05	0.95	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
0.025	0.975		6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06
0.01	0.99		8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80
0.05	0.95	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
0.025	0.975		5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77
0.01	0.99		8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37
0.05	0.95	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
0.025	0.975		5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64
0.01	0.99		7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17
0.05	0.95	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
0.025	0.975		5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51
0.01	0.99		7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98
0.05	0.95	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
0.025	0.975		5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39
0.01	0.99		7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80
0.05	0.95	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
0.025	0.975		5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27
0.01	0.99		7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63
0.05	0.95	120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91
0.025	0.975		5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16
0.01	0.99		6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47
0.05	0.95	∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83
0.025	0.975		5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05
0.01	0.99		6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32

Table VII *continued*

$$P(F \le f) = \int_0^f \frac{\Gamma[(r_1 + r_2)/2](r_1/r_2)^{r_1/2} w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)(1 + r_1 w/r_2)^{(r_1+r_2)/2}}\, dw$$

α	$P(F \le f)$	Den. d.f. r_2	Numerator Degrees of Freedom, r_1								
			12	15	20	24	30	40	60	120	∞
0.05	0.95	1	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
0.025	0.975		976.71	984.87	993.10	997.25	1001.4	1005.6	1009.8	1014.0	1018.3
0.01	0.99		6106	6157	6209	6235	6261	6287	6313	6339	6366
0.05	0.95	2	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
0.025	0.975		39.42	39.43	39.45	39.46	39.47	39.47	39.48	39.49	39.50
0.01	0.99		99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
0.05	0.95	3	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
0.025	0.975		14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90
0.01	0.99		27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13
0.05	0.95	4	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
0.025	0.975		8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26
0.01	0.99		14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
0.05	0.95	5	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
0.025	0.975		6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02
0.01	0.99		9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
0.05	0.95	6	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
0.025	0.975		5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85
0.01	0.99		7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
0.05	0.95	7	3.57	3.51	3.41	3.41	3.38	3.34	3.30	3.27	3.23
0.025	0.975		4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14
0.01	0.99		6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
0.05	0.95	8	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
0.025	0.975		4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67
0.01	0.99		5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
0.05	0.95	9	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
0.025	0.975		3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33
0.01	0.99		5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31

Table VII *continued*

$$P(F \le f) = \int_0^f \frac{\Gamma[(r_1 + r_2)/2](r_1/r_2)^{r_1/2} w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)(1 + r_1 w/r_2)^{(r_1+r_2)/2}}\, dw$$

α	$P(F \le f)$	Den. d.f. r_2	Numerator Degrees of Freedom, r_1								
			12	15	20	24	30	40	60	120	∞
0.05	0.95	10	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
0.025	0.975		3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.08
0.01	0.99		4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
0.05	0.95	12	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
0.025	0.975		3.28	3.18	3.07	3.02	2.96	2.91	2.85	2.79	2.72
0.01	0.99		4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
0.05	0.95	15	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
0.025	0.975		2.96	2.86	2.76	2.70	2.64	2.59	2.52	2.46	2.40
0.01	0.99		3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
0.05	0.95	20	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
0.025	0.975		2.68	2.57	2.46	2.41	2.35	2.29	2.22	2.16	2.09
0.01	0.99		3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
0.05	0.95	24	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
0.025	0.975		2.54	2.44	2.33	2.27	2.21	2.15	2.08	2.01	1.94
0.01	0.99		3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
0.05	0.95	30	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
0.025	0.975		2.41	2.31	2.20	2.14	2.07	2.01	1.94	1.87	1.79
0.01	0.99		2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
0.05	0.95	40	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
0.025	0.975		2.29	2.18	2.07	2.01	1.94	1.88	1.80	1.72	1.64
0.01	0.99		2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
0.05	0.95	60	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
0.025	0.975		2.17	2.06	1.94	1.88	1.82	1.74	1.67	1.58	1.48
0.01	0.99		2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
0.05	0.95	120	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
0.025	0.975		2.05	1.95	1.82	1.76	1.69	1.61	1.53	1.43	1.31
0.01	0.99		2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
0.05	0.95	∞	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00
0.025	0.975		1.94	1.83	1.71	1.64	1.57	1.48	1.39	1.27	1.00
0.01	0.99		2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

I Multiple Choices (28 points)

- **4 points for each correct answer; 0 point for each incorrect answer or no answer.**
- **For each question, only choose (at most) one out of four given choices (A,B,C and D). If you choose more than one choice in one question, your answer will be incorrect.**

1. (4 points) Suppose $X_1, X_2, \ldots, X_n$ are i.i.d. random variables with probability density function (pdf)

$$f(x;\sigma) = \frac{1}{2\sigma}\exp(-\frac{|x|}{\sigma}), \quad x \in \mathbb{R}.$$

Which of the following is the maximum likelihood estimator of σ?

A. $\widehat{\sigma} = \dfrac{\sum_{i=1}^n |X_i|}{n}$ B. $\widehat{\sigma} = \dfrac{\sum_{i=1}^n |X_i|}{n-1}$

C. $\widehat{\sigma} = \sqrt{\dfrac{\sum_{i=1}^n X_i^2}{2n}}$ D. $\widehat{\sigma} = \sqrt{\dfrac{\sum_{i=1}^n X_i^2}{2n-2}}$

Solution: (A)
The log-likelihood function is $l(\sigma) = \sum_{i=1}^n \left[-\log 2 - \log\sigma - \frac{|X_i|}{\sigma}\right]$. By taking the derivative with respect to σ be zero, this gives us the MLE for σ is $\hat{\sigma} = \dfrac{\sum_{i=1}^n |X_i|}{n}$.

2. (4 points) Suppose that $X_1, X_2, \ldots, X_n$ form a random sample from the normal distribution with mean μ and variance σ^2 unknown. To make $\widehat{\sigma}^2 = k\sum_{i=1}^{n-1}(X_{i+1} - X_i)^2$ an unbiased estimator of σ^2, which value should the constant k equal to?

A. $\dfrac{1}{n-1}$ B. $\dfrac{1}{n}$ C. $\dfrac{1}{2(n-1)}$ D. $\dfrac{1}{2n}$

Solution: (C)
Since

$$\begin{aligned} E[(X_{i+1} - X_i)^2] &= E(X_{i+1}^2) - 2E(X_{i+1})E(X_i) + E(X_i^2) \\ &= \sigma^2 + \mu^2 - 2\mu^2 + \sigma^2 + \mu^2 \\ &= 2\sigma^2, \end{aligned}$$

we have

$$E(\widehat{\sigma}^2) = k \sum_{i=1}^{n-1} 2\sigma^2 = 2k(n-1)\sigma^2.$$

By setting $E(\widehat{\sigma}^2) = \sigma^2$, we obtain $k = \dfrac{1}{2(n-1)}$.

3. (4 points) A bakery claims that its cookies weigh 100 grams. Michael tests this claim using a two-sided hypothesis test. He buys 25 cookies, and obtains a sample mean of $\bar{x} = 95$ g, and a sample standard deviation of $s = 10$ g. Assuming that the weights of cookies are i.i.d. normal random variables, Michael's p–value is approximately

A. 0.0062
B. 0.01
C. 0.0124
D. 0.02

Solution: (D)
The realized test statistic is

$$t = \frac{95 - 100}{10/\sqrt{25}} = -\frac{5}{2} = -2.5.$$

The p–value is

$$2P(t(24) > |t|) = 2P(t(24) > 2.5) \approx 2 \times 0.01 = 0.02.$$

4. (4 points) Recall the simple linear regression model we learnt in class:

$$Y = \alpha + \beta(x - \overline{x}) + \epsilon.$$

Given the following three data $\{(x_i, y_i), i = 1, 2, 3\}$, which one is the correct 95% confidence interval for α?

x_i	y_i
1.0	1.0
2.0	2.0
3.0	1.3

A. $[-3.66, 6.52]$ B. $[-2.16, 5.03]$
C. $[-1.51, 4.37]$ D. $[-2.97, 5.84]$

Solution: (A)

$$\hat{\alpha} = \frac{\sum y_i}{n} = 1.43, \quad \hat{\beta} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = 0.15$$

$$S_R = \sqrt{\frac{1}{n-2}\sum \left(y_i - \hat{\alpha} - \hat{\beta}(x_i - \bar{x})\right)^2} = 0.694$$

$$t_{\alpha/2}(n-2) = t_{0.025}(1) = 12.706$$

$$\hat{\alpha} \pm t_{\alpha/2}(n-2) S_R \sqrt{\frac{1}{n}} = 1.43 \pm 12.706 * 0.694 \sqrt{\frac{1}{3}} = [-3.66, 6.52]$$

5. (4 points) Based on the previous question, which one is the 95% prediction interval for Y_4 given a new data $x_4 = 4$?
A. $[-11.494, 14.961]$ B. $[-14.164, 17.631]$
C. $[-14.367, 17.833]$ D. $[-11.736, 15.203]$

Solution: (C)

A 95% prediction interval for Y_4 is

$$\hat{Y}_4 = \hat{\alpha} + \hat{\beta}(x_4 - \bar{x}) = 1.43 + 0.15 * (4 - 2) = 1.73$$

$$\hat{\alpha} + \hat{\beta}(x_4 - \bar{x}) \pm t_{0.025}(3-2) S_R \sqrt{1 + \frac{1}{3} + \frac{(x_4 - \bar{x})^2}{\sum_{i=1}^{3}(x_i - \bar{x})^2}}$$

$$= 1.73 \pm 12.706 * 0.694 \sqrt{1 + \frac{1}{3} + \frac{(4-2)^2}{2}} = [-14.367, 17.833]$$

6. (4 points) Let $X_{(1)} < X_{(2)} < \ldots < X_{(5)}$ be the order statistics of five i.i.d. observations from a distribution with pdf

$$f(x) = \begin{cases} 3x^2, & \text{for } 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find $P(X_{(2)} < \frac{1}{2})$.

A. 0.2367 B. 0.1207 C. 0.1468 D. 0.2879

Solution: (B)
The cdf of X_i is

$$P(X_i < y) = y^3$$

The pdf of $X_{(2)}$ is

$$\begin{aligned} g(y) &= \frac{5!}{(2-1)!(5-2)!}(F(y))^{2-1} f(y)(1 - F(y))^{5-2} \\ &= 20 \cdot y^3 \cdot 3y^2 \cdot (1 - y^3)^3 \\ &= 60y^5(1 - y^3)^3, \qquad 0 < y < 1 \end{aligned}$$

Then

$$\begin{aligned} P\left(X_{(2)} < \frac{1}{2}\right) &= \int_0^{1/2} 60y^5(1 - y^3)^3 dy \\ &= \int_0^{1/8} 20z(1 - z)^3 dz = 0.1207 \end{aligned}$$

7. (4 points) Let p denote the probability that, for a particular tennis player, the first serve is good. Since $p = 0.40$, this player decided to take lessons in order to increase p. When the lessons are completed, the hypothesis $H_0 : p = 0.4$ will be tested against $H_1 : p > 0.4$ on the basis of $n = 25$ trials. Let Y equal the number of first serves that are good, and let the critical region be defined by $C = \{y : y \geq 13\}$. What is the probability of type II error if the true value of p after the lessons is $p = 0.6$?

   A. 0.1103 B. 0.1538 C. 0.0888 D. 0.2677

Solution: (B)

$$\begin{aligned}\beta &= P(Y < 13; p = 0.6) = P(Y \leq 12; p = 0.6) \\ &= P(25 - Y \geq 25 - 12) \qquad \text{where } 25 - Y \sim Bin(25, 0.4) \\ &= 1 - 0.8462 = 0.1538\end{aligned}$$

II Regular Questions (72 points)

8. (12 points) Michael likes to play basketball. He attempts 100 free throws, and makes 70 of them. Assume that the outcomes of his free throws are independent of one another.

   (a) (4 points) Compute a 95% confidence interval for p_M, his true free throw percentage.

   (b) (4 points) Before attempting these shots, he boasted to his friend Charles that he makes free throws 80% of the time. Charles was skeptical, and conducted the hypothesis test

   $$H_0 : p_M = 0.8 \qquad H_1 : p_M \neq 0.8.$$

   at the $\alpha = 0.05$ significance level. Does Charles reject H_0?

   (c) (4 points) After Michael finished, his rival, Tom, shot 100 free throws of his own, and made 90 of them. Letting p_T denote Tom's true free throw percentage, compute a 95% confidence interval for $p_M - p_T$. Assume independence.

Solution: Part (a).

$$\widehat{p}_M = 0.7, \quad z_{0.025} = 1.96.$$

The 95% C.I. is

$$0.7 \pm 1.96\sqrt{\frac{(0.7)(0.3)}{100}} = [0.6102, 0.7898].$$

Part (b). The realized test statistic is

$$t = \frac{0.7 - 0.8}{\sqrt{\frac{0.8(0.2)}{100}}} = -2.5$$

Since

$$|t| = 2.5 > 1.96 = z_{0.025},$$

Charles rejects H_0 at 5%.

Part (c).

$$\widehat{p}_M = 0.7, \quad \widehat{p}_T = 0.9, \quad z_{0.025} = 1.96.$$

The 95% C.I. is

$$0.7 - 0.9 \pm 1.96\sqrt{\frac{(0.7)(0.3)}{100} + \frac{(0.9)(0.1)}{100}} = [-0.3074, -0.0926].$$

9. (12 points) There are many couples in CUHKSZ. Let X and Y represent the height of boyfriend and girlfriend, respectively. Assume that the distribution of $X_1, \ldots, X_n \overset{i.i.d}{\sim} N(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \overset{i.i.d}{\sim} N(\mu_Y, \sigma_Y^2)$. We are curious about the difference between the height of boyfriends and the height of girlfriends. A random sample test of eight couples on campus obtained the following data: $\bar{x} = 175, \bar{y} = 165, s_x^2 = 80, s_y^2 = 48$. You need to calculate the two-sided 95% confidence interval for the difference $\mu_X - \mu_Y$ under different assumptions.

   (a) (4 points) Suppose that $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ are independent, and they share the same but unknown variance. Calculate the two-sided 95% confidence interval for the difference $\mu_X - \mu_Y$.

   (b) (4 points) Suppose that $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ are indepen-

dent, and they have different (and unknown) variances. Calculate the two-sided 95% confidence interval for the difference $\mu_X - \mu_Y$.

(c) (4 points) Suppose that $n = m$ and X_i, Y_i are dependent for $i = 1, \ldots, n$. However, the n pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$ are independent. Let $D_i = X_i - Y_i$, and we know that the sample variance of D_i is $s_D^2 = 7^2$. Calculate the two-sided 95% confidence interval for the difference $\mu_X - \mu_Y$.

Solution: Part (a) Since $n = m = 8$, and $\sigma_X^2 = \sigma_Y^2$ unknown, we have 95% C.I. to be

$$\bar{X} - \bar{Y} \pm t_{\alpha/2}(n + m - 2) S_p \sqrt{\frac{1}{n} + \frac{1}{m}}$$

$$t_{0.025}(14) = 2.145, \quad \sqrt{\frac{1}{n} + \frac{1}{m}} = \sqrt{\frac{1}{8} + \frac{1}{8}} = 0.5$$

$$S_p = \sqrt{\frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}} = \sqrt{\frac{7(80) + 7(48)}{14}} = \sqrt{\frac{128}{2}} = 8$$

$$175 - 165 \pm 2.145 * 8 * 0.5 = [1.42, 18.58]$$

Part (b) Since $n = m = 8$, and $\sigma_X^2 \neq \sigma_Y^2$ unknown, we have 95% C.I. to be

$$\bar{X} - \bar{Y} \pm t_{\alpha/2}(r) \sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor = \left\lfloor \frac{16^2}{\frac{80^2 + 48^2}{7 * 8^2}} \right\rfloor = 13$$

$$t_{0.025}(13) = 2.160, \quad \sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}} = \sqrt{\frac{80 + 48}{8}} = 4$$

$$175 - 165 \pm 2.160 * 4 = [1.36, 18.64]$$

Part (c) Since the n pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$ are independent,

let $D_i = X_i - Y_i$. Then the 95% CI is

$$\bar{D} \pm t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}$$

$$\bar{D} = \bar{X} - \bar{Y} = 175 - 165 = 10, \quad t_{0.025}(7) = 2.365$$

$$10 \pm 2.365\frac{7}{\sqrt{8}} = [4.146, 15.853]$$

10. (12 points) Students looked at the effect of a certain fertilizer on plant growth. The students tested this fertilizer on one group of plants (Group A) and did not give fertilizer to second group (Group B). Assume that the distribution of the growths of the plants in group A and B are $N(\mu_A, \sigma_A^2)$, and $N(\mu_B, \sigma_B^2)$, and they are independent. The growths of the plants, in mm, over six weeks were as follow:

Group A:	55	61	33	57	17	46	50	42	71	51	63
Group B:	31	27	12	44	9	25	34	53	33	21	32

(a) (4 points) Conduct a two-sided F-test on

$$H_0 : \sigma_A^2 = \sigma_B^2, \quad H_1 : \sigma_A^2 \neq \sigma_B^2.$$

at the significance level $\alpha = 0.05$. State your conclusion.

(b) (4 points) Based on the result of part (a), find the critical region for testing $H_0 : \mu_A = \mu_B$ against $H_1 : \mu_A > \mu_B$ at the significance level $\alpha = 0.05$. State your conclusion.

(c) (4 points) Student Tom believed that fertilizer has a significant influence on plant growth according to past experience. Therefore he wants to test $H_0 : \mu_A = 2\mu_B$ against $H_1 : \mu_A < 2\mu_B$ at the significance level $\alpha = 0.05$. Assume that $\sigma_A^2 = 10^2$, $\sigma_B^2 = 13^2$. (Hint: what is the test statistic?)

Solution:

(a)

$$0.2688 = F_{0.975}(10, 10) < \frac{S_A^2}{S_B^2} = \frac{226.2545}{162.7636} = 1.39 < F_{0.025}(10, 10) = 3.72$$

Therefore, we do not reject null hypothesis, the two variances are equal.

(b) Based on the result of part (a), we should use the pooled t-test to test $H_0 : \mu_A = \mu_B$ against $H_1 : \mu_A > \mu_B$.

$$T = \frac{\bar{A} - \bar{B}}{s_p\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} = \frac{49.6363 - 29.1818}{13.9466\sqrt{\frac{1}{11} + \frac{1}{11}}} = 3.4395 > t_{0.05}(20) = 1.725$$

Therefore we reject $H_0 : \mu_A = \mu_B$.

(c) Since $A \sim N(\mu_A, 10^2)$, $B \sim N(\mu_B, 13^2)$, and they are independent. Then

$$\bar{A} \sim N(\mu_A, \frac{10^2}{11}), \bar{B} \sim N(\mu_B, \frac{13^2}{11})$$
$$\bar{A} - 2\bar{B} \sim N(\mu_A - 2\mu_B, \frac{10^2}{11} + \frac{4 \times 13^2}{11})$$

Therefore the critical region should be

$$\frac{\bar{A} - 2\bar{B} - (\mu_A - 2\mu_B)}{\sqrt{\frac{10^2}{11} + \frac{4(13^2)}{11}}} \leq -z_{0.05}$$

However

$$\frac{\bar{A} - 2\bar{B} - (\mu_A - 2\mu_B)}{\sqrt{\frac{10^2}{11} + \frac{4(13^2)}{11}}} = -1.039 > -1.645$$

Therefore we do not reject $H_0 : \mu_A = 2\mu_B$.

11. (20 points) (a) Suppose that $X_1, \ldots, X_n$ are i.i.d. random variables with pdf given by

$$f(x;\theta) = \begin{cases} \theta x^{\theta-1}, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a1) (5 points) Find the maximum likelihood estimator $\widehat{\theta}$ of θ.

(a2) (3 points) Suppose the following $n = 10$ samples is generated from this distribution with parameter θ:

0.77 0.71 0.90 0.89 0.33 0.69 0.19 0.03 0.74 0.50

Compute the maximum likelihood estimate $\widehat{\theta}$.

(b) Suppose now $Y_1, \ldots, Y_n$ are i.i.d. random samples from the uniform distribution on the interval $[\theta_1, \theta_2]$, where both θ_1 and θ_2 are unknown.

(b1) (8 points) Find the maximum likelihood estimators $\widehat{\theta}_1$ and $\widehat{\theta}_2$ of θ_1 and θ_2 respectively.

(b2) (4 points) Find the pdf of $\widehat{\theta}_1$ and $\widehat{\theta}_2$. State clearly the support of them in your answer.

Solution:

Part (a1).

The log-likelihood function is $l(\theta) = n \ln \theta + (\theta - 1) \sum_{i=1}^{n} \ln x_i$, let the derivative with respect to θ be zero, we have $\frac{n}{\theta} + \sum_{i=1}^{n} \ln x_i = 0$, which gives us the M.L.E. for θ as

$$\widehat{\theta} = -\frac{n}{\sum_{i=1}^{n} \ln X_i}.$$

Part (a2).

$$\widehat{\theta} = -\frac{10}{\ln \prod_{i=1}^{10} x_i} = 1.1811.$$

Part (b1).

The log-likelihood function is $l(\theta_1, \theta_2) = -n \log(\theta_2 - \theta_1)$, compute the derivative with respect to θ_1 and θ_2, we have

$$\frac{\partial l(\theta_1, \theta_2)}{\partial \theta_1} = \frac{n}{\theta_2 - \theta_1} > 0$$
$$\frac{\partial l(\theta_1, \theta_2)}{\partial \theta_2} = -\frac{n}{\theta_2 - \theta_1} < 0$$

Therefore, we can get the maximum of the log-likelihood $l(\theta_1, \theta_2)$ when $\theta_1 = \min(x_1, \ldots, x_n)$ and $\theta_2 = \max(x_1, \ldots, x_n)$, which gives us the M.L.E.'s for θ_1 and θ_2 as

$$\widehat{\theta}_1 = \min(Y_1, \ldots, Y_n) = Y_{(1)}.$$
$$\widehat{\theta}_2 = \max(Y_1, \ldots, Y_n) = Y_{(n)}.$$

Part (b2).
Let $Y_{(1)}$ be $\min\{Y_1, \ldots, Y_n\}$ and $Y_{(n)}$ be $\max\{Y_1, \ldots, Y_n\}$. The pdf of $\widehat{\theta}_1$ is

$$f_{Y_{(1)}}(x) = \frac{n(\theta_2 - x)^{n-1}}{(\theta_2 - \theta_1)^n} \mathbf{1}_{[\theta_1, \theta_2]}(x).$$

and the pdf of $\widehat{\theta}_2$ is

$$f_{Y_{(n)}}(x) = \frac{n(x - \theta_1)^{n-1}}{(\theta_2 - \theta_1)^n} \mathbf{1}_{[\theta_1, \theta_2]}(x).$$

12. (16 points) Consider the linear regression model that we learnt in class:

$$Y_i = \alpha + \beta(x_i - \overline{x}) + \epsilon_i, \quad i = 1, 2, \ldots, n,$$

where x_i are fixed and $\epsilon_1, \ldots, \epsilon_n \overset{i.i.d.}{\sim} N(0, \sigma^2)$ are the random noise. Denote by $\widehat{\alpha}$ and $\widehat{\beta}$ to be the maximum likelihood estimator of α and β respectively. For $i = 1, 2, \ldots, n$, define

$$\widehat{Y}_i = \widehat{\alpha} + \widehat{\beta}(x_i - \overline{x}),$$
$$\widehat{\epsilon}_i = Y_i - \widehat{Y}_i.$$

(a) (5 points) Prove that

$$\frac{1}{n} \sum_{i=1}^{n} \widehat{Y}_i = \overline{Y},$$

where $\overline{Y}$ is the sample mean of Y_i with $i = 1, \ldots, n$.

(b) (2 points) Prove that

$$\sum_{i=1}^{n} \widehat{\epsilon}_i = 0.$$

(c) (7 points) Prove that

$$\sum_{i=1}^{n} \widehat{Y}_i \widehat{\epsilon}_i = 0.$$

(d) (2 points) Prove that

$$\sum_{i=1}^{n} Y_i^2 = \sum_{i=1}^{n} \widehat{Y}_i^2 + \sum_{i=1}^{n} \widehat{\epsilon}_i^2.$$

Solution: First, recall that

$$\widehat{\alpha} = \overline{Y}$$
$$\widehat{\beta} = \frac{\sum_{i=1}^{n} Y_i \, (x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2}$$

Part (a).

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^{n} \widehat{Y}_i &= \widehat{\alpha} + \frac{\widehat{\beta}}{n} \sum_{i=1}^{n} (x_i - \overline{x}) \\ &= \overline{Y} + 0 = \overline{Y}. \end{aligned}$$

Part (b). We can either check by definition or use part (a). Using part (a),

$$\sum_{i=1}^{n} \widehat{\epsilon}_i = n\overline{Y} - \sum_{i=1}^{n} \widehat{Y}_i = 0.$$

Part (c). Using part (b), we have

$$\begin{aligned} \sum_{i=1}^{n} \widehat{Y}_i \widehat{\epsilon}_i &= \sum_{i=1}^{n} \left(\widehat{\alpha} + \widehat{\beta}(x_i - \overline{x}) \right) \widehat{\epsilon}_i \\ &= \widehat{\alpha} \sum_{i=1}^{n} \widehat{\epsilon}_i + \widehat{\beta} \sum_{i=1}^{n} x_i \widehat{\epsilon}_i - \widehat{\beta}\overline{x} \sum_{i=1}^{n} \widehat{\epsilon}_i \\ &= \widehat{\beta} \sum_{i=1}^{n} x_i \widehat{\epsilon}_i \\ &= \widehat{\beta} \sum_{i=1}^{n} x_i (Y_i - \overline{Y} - \widehat{\beta}(x_i - \overline{x})) \\ &= \widehat{\beta} \sum_{i=1}^{n} x_i (Y_i - \overline{Y}) - \widehat{\beta}^2 \sum_{i=1}^{n} x_i (x_i - \overline{x}) \\ &= \widehat{\beta} \sum_{i=1}^{n} Y_i (x_i - \overline{x}) - \widehat{\beta}^2 \sum_{i=1}^{n} (x_i - \overline{x})^2 \\ &= 0. \end{aligned}$$

Part (d). Consider

$$Y_i^2 = (\widehat{Y}_i + \widehat{\epsilon}_i)^2 = \widehat{Y}_i^2 + 2\widehat{Y}_i\widehat{\epsilon}_i + \widehat{\epsilon}_i^2.$$

Summing over i from 1 to n and using part (c), we have the desired result.